

INFRASTRUCTURE AS A SERVICE



Work Integrated Learning Programmes

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AGENDA

IaaS Recap

AWS



Work Integrated Learning Programmes

WHAT IS INFRASTRUCTURE AS A SERVICE?

Resources for the Computing and Storage are delivered as a Service

Maximum Flexibility For Users

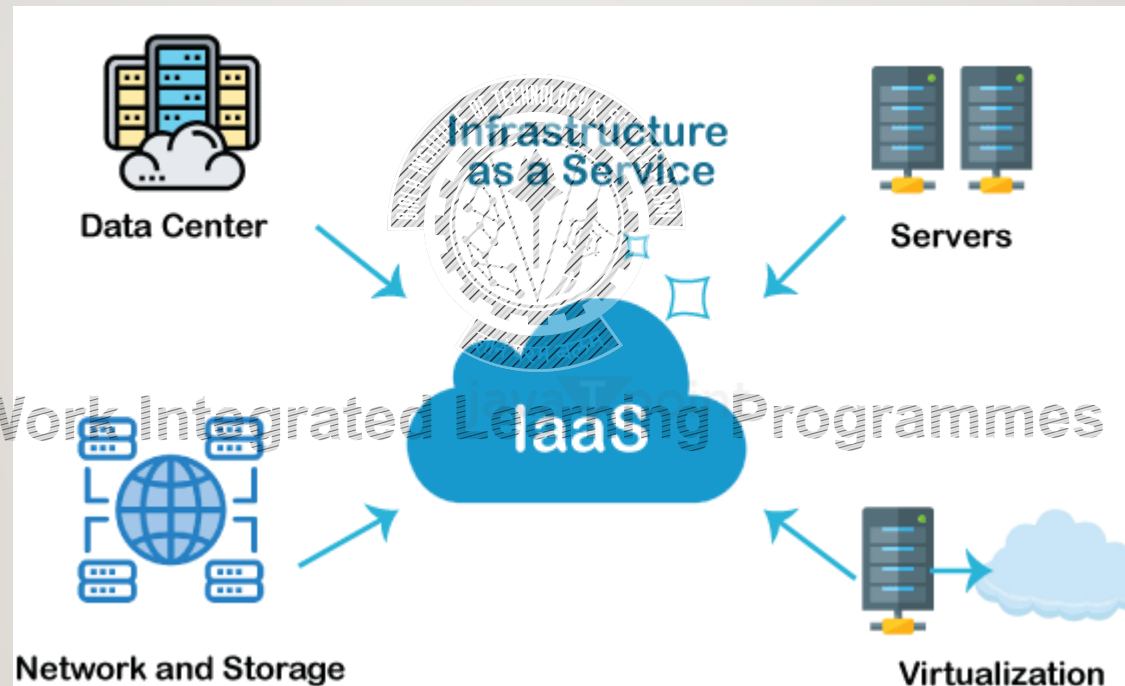
Virtual Computing and the usage of Storage is left to the cloud users

Users can load the needed OS and Software

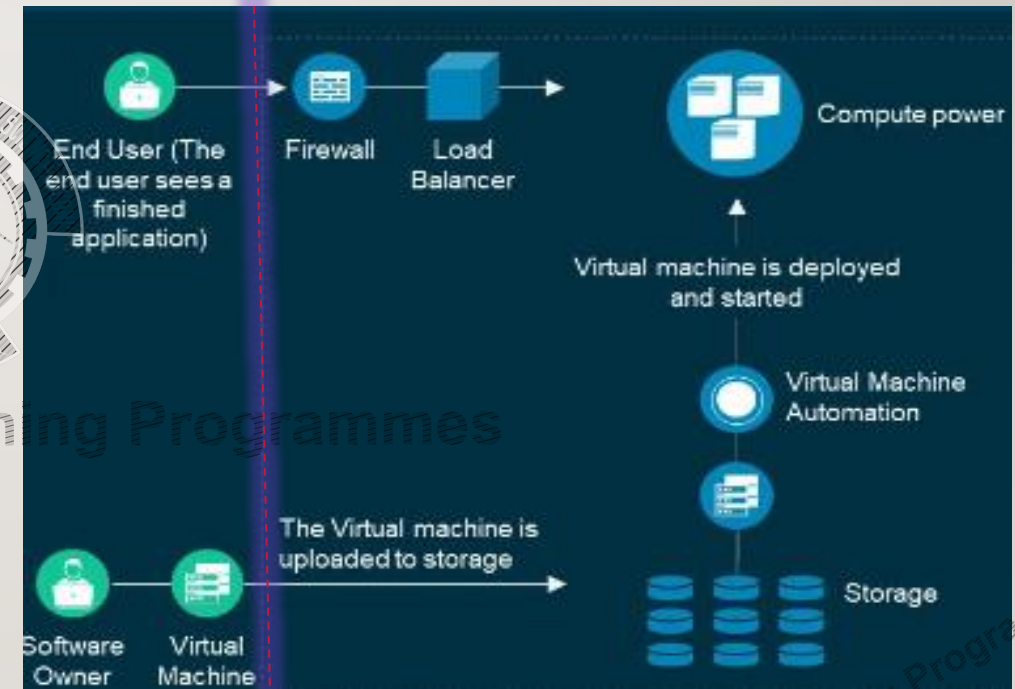
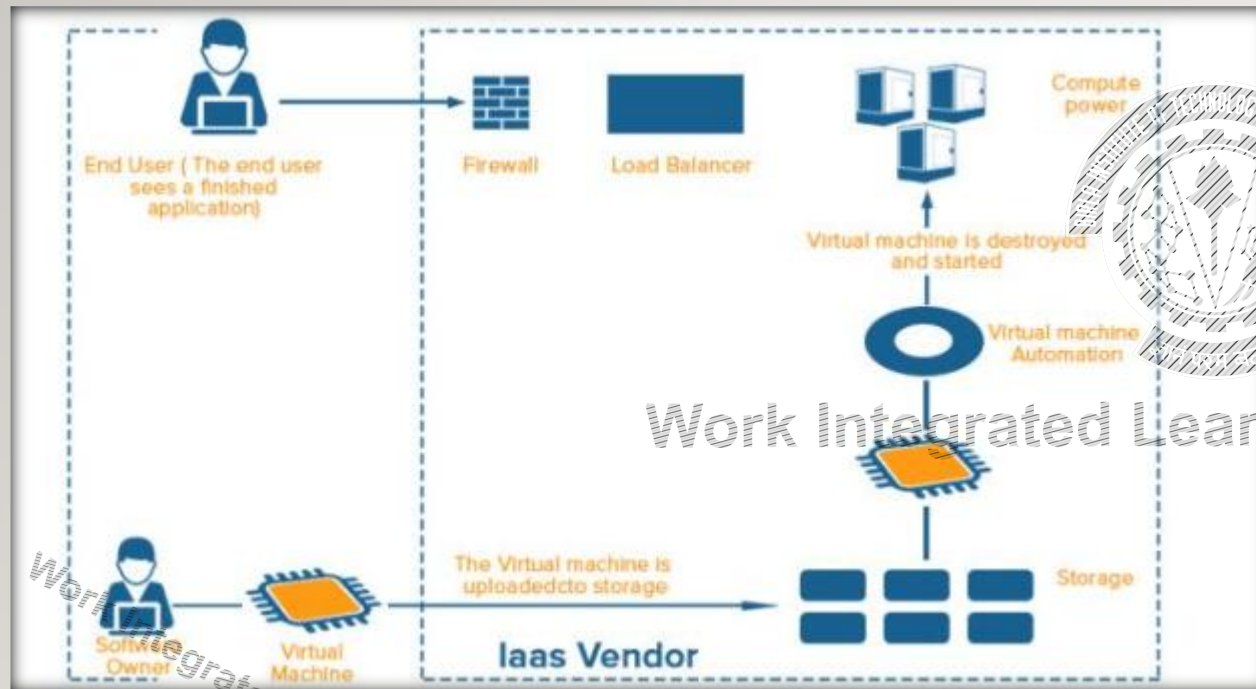
Burden of maintenance is definitely with



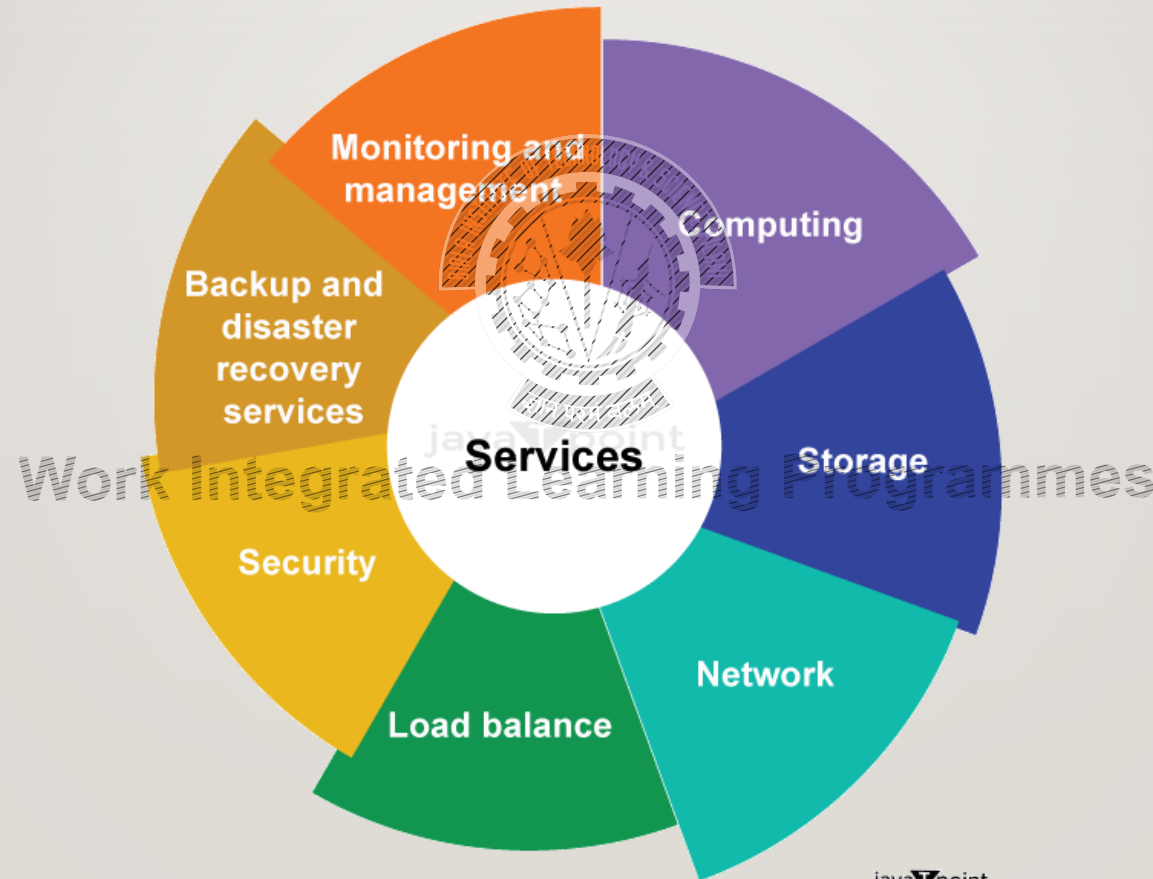
WHAT IS INFRASTRUCTURE AS A SERVICE?



HOW IAAS WORKS?



SERVICES



BENEFITS

- **Scalability:** Enables users to adjust computing capacity according to their demands without requiring long lead times or up-front hardware purchases.
- **Virtualization:** Uses virtualization technology to generate virtualized instances that can be managed and delivered on-demand by abstracting physical computer resources.
- **Resource Pooling:** This feature enables users to share computer resources, such as networking and storage, among a number of users, maximizing resource utilization and cutting costs.
- **Elasticity:** Allows users to dynamically modify their computing resources in response to shifting demand, ensuring optimum performance and financial viability.
- **Self-Service:** Offers "self-service" portals that let them independently deploy, administer and monitor their computing resources without the assistance of IT employees.
- **Availability:** To ensure the high availability and reliability of services, IaaS providers often run redundant and geographically dispersed data centers.
- **Security:** To safeguard their infrastructure and client data, IaaS companies adopt security measures, including data encryption, firewalls, access controls, and threat detection.
- **Customization:** Enables users to alter the operating systems, application stacks & security settings of their virtualized instances to suit their unique requirements.

WHAT SHOULD CUSTOMERS DO – SECURITY PERSPECTIVE



SHARED RESPONSIBILITY MODEL

Let's take the example for Patches - Cloud Provider AWS.

Security and Compliance is a shared responsibility between AWS and the customer.

In AWS, the responsibility for patching depends on the type of service. For virtual machines (VMs) like EC2 instances, the responsibility follows the Shared Responsibility Model

AWS Responsibility: AWS is responsible for the security "of" the cloud, which includes the infrastructure, hardware, software, networking, and facilities that run AWS Cloud services

Customer Responsibility: The customers are responsible for the security "in" the cloud. This includes managing and patching the guest operating system (Windows or Linux) on their EC2 instances, any application software or utilities installed on the instances, and the configuration of the AWS-provided firewall (security groups)

- Responsibility
- Who does what

<https://aws.amazon.com/compliance/shared-responsibility-model/https://www.impactmakers.com/verify-ec2-instance-patching-in-aws/>

Shared Responsibility Model

IaaS

Physical
Infrastructure
Network
Virtualization
Operating System
Application
Data
Service Configuration

PaaS

Physical
Infrastructure
Network
Virtualization
Operating System
Application
Data
Service Configuration

SaaS

Physical
Infrastructure
Network
Virtualization
Operating System
Application
Data
Service Configuration



Customer Specific Responsibility

CONTROL

Inherited Controls – Controls which a customer fully inherits from AWS.

- Physical and Environmental controls

Shared Controls – Controls which apply to both the infrastructure layer and customer layers, but in completely separate contexts or perspectives. In a shared control, AWS provides the requirements for the infrastructure and the customer must provide their own control implementation within their use of AWS services. Examples include:

- Patch Management – AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest OS and applications.
- Configuration Management – AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.
- Awareness & Training - AWS trains AWS employees, but a customer must train their own employees.

Customer Specific – Controls which are solely the responsibility of the customer based on the application they are deploying within AWS services. Examples include:

- Service and Communications Protection or Zone Security which may require a customer to route or zone data within specific security environments.

CUSTOMER

RESPONSIBILITY FOR
SECURITY 'IN' THE CLOUD

AWS

RESPONSIBILITY FOR
SECURITY 'OF' THE CLOUD

CUSTOMER DATA

PLATFORM, APPLICATIONS, IDENTITY & ACCESS MANAGEMENT

OPERATING SYSTEM, NETWORK & FIREWALL CONFIGURATION

CLIENT-SIDE DATA
ENCRYPTION & DATA INTEGRITY
AUTHENTICATION

SERVER-SIDE ENCRYPTION
(FILE SYSTEM AND/OR DATA)

NETWORKING TRAFFIC
PROTECTION (ENCRYPTION,
INTEGRITY, IDENTITY)

SOFTWARE

COMPUTE

STORAGE

DATABASE

NETWORKING

HARDWARE/AWS GLOBAL INFRASTRUCTURE

REGIONS

AVAILABILITY ZONES

EDGE LOCATIONS

•**Security.** While the customer is in control of the applications, data, middleware, and the operating system platform, security threats can still be sourced from the host or other virtual machines (VMs). Insider threat or system vulnerabilities may expose data communication between the host infrastructure and VMs to unauthorized entities.

•**Legacy systems operating in the cloud.** While customers can run legacy applications in the cloud, the infrastructure may not be designed to deliver specific controls to secure them. Minor enhancements to legacy applications may be required before migrating them to the cloud, possibly leading to new security issues unless adequately tested for security and performance in the IaaS systems.

•**Internal resources and training.** Additional resources and training may be required for the workforce to learn how to effectively manage the infrastructure. Customers will be responsible for data security, backup, and business continuity. Due to inadequate control of the infrastructure, however, monitoring and management of the resources may be difficult without availability of adequate in-house training and resources. Vendors have custom trainings and there are also training agencies / institutes.

•**Multi-tenant security.** Since the hardware resources are dynamically allocated across users as needed, the vendor is required to ensure that other customers cannot access data left on storage assets by previous customers. Similarly, customers must rely on the vendor to ensure that VMs are adequately isolated within the multi-tenant cloud architecture. Let us say that a customer runs two projects (for two different clients) and even for those VMs, the isolation is needed as the clients of the customer would definitely want so.



Determine external and internal security and related compliance requirements and objectives, and consider industry frameworks like the [NIST Cybersecurity Framework \(CSF\)](#) and [ISO](#).



Consider employing the [AWS Cloud Adoption Framework \(CAF\)](#) and [Well-Architected best practices](#) to plan and execute your digital transformation at scale.



Review the security functionality and configuration options of individual AWS services within the security chapters of [AWS service documentation](#).



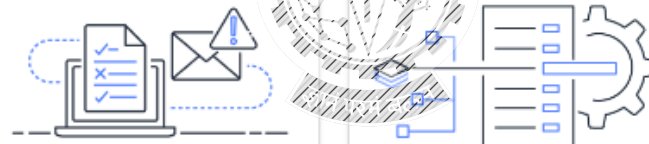
Evaluate the [AWS Security, Identity, and Compliance services](#) to understand how they can be used to help meet your security and compliance objectives.



Review [third-party audit attestation documents](#) to determine inherited controls and what required controls may be remaining for you to implement in your environment.



Provide your internal and external audit teams with cloud-specific learning opportunities by leveraging the [Cloud Audit Academy](#) training programs.



Perform a [Well-Architected Review](#) of your AWS workloads to evaluate the implementation of best practices for security, reliability, and performance.



Explore solutions available in the [AWS Marketplace](#) digital catalog with thousands of software listings from independent software vendors that enable you to find, test, buy, and deploy software that runs on AWS.



Explore [AWS Security Competency Partners](#) offering expertise and proven customer success securing every stage of cloud adoption, from initial migration through ongoing day-to-day management.

VENDORS...



Many others...

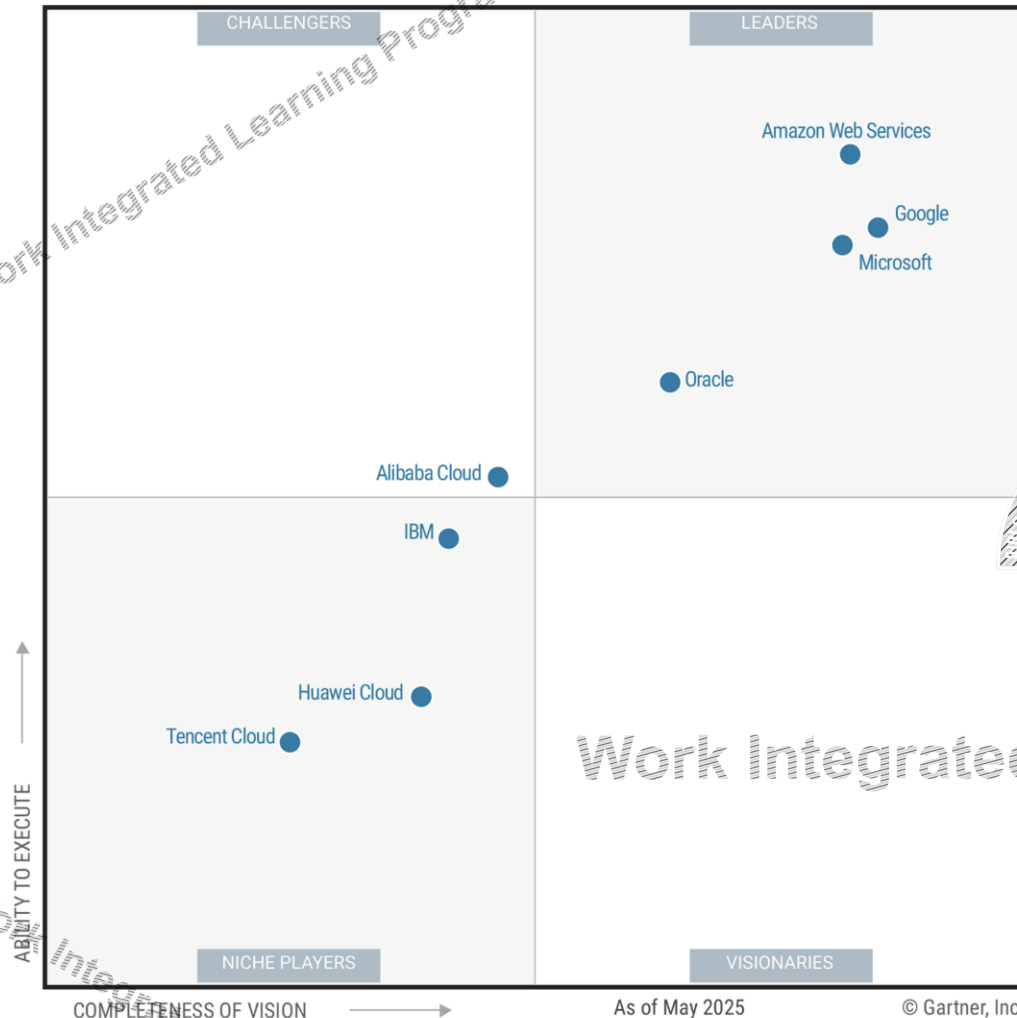
- [DigitalOcean](#)
- [Linode](#)
- [Rackspace](#)
- [Cisco Metacloud](#)

<https://www.vpsbenchmarks.com/hosters>

(51 Providers and counting....)

<https://www.vpsbenchmarks.com/hosters>

Figure 1: Magic Quadrant for Strategic Cloud Platform Services

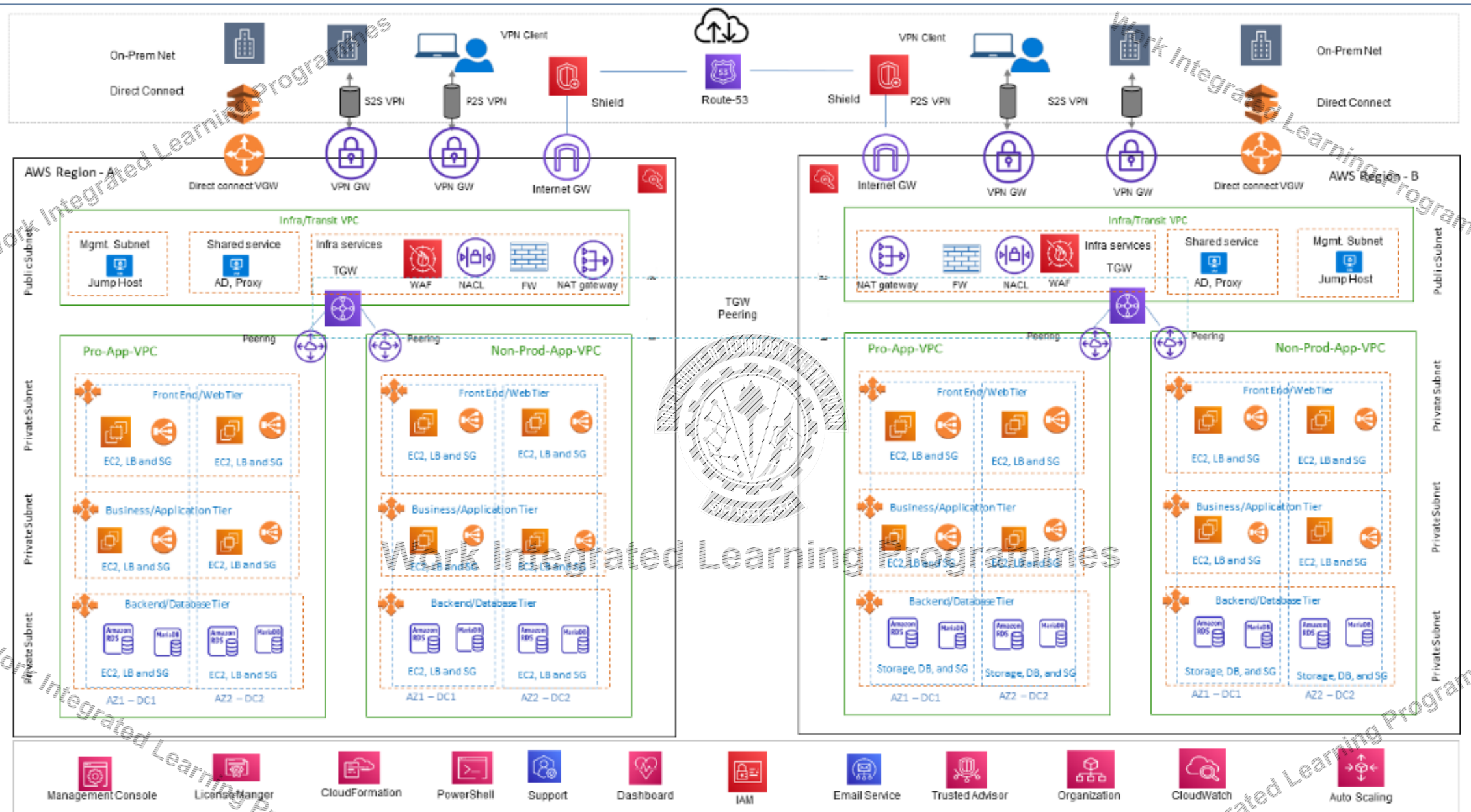


AWS named as a Leader in 2025 Gartner Magic Quadrant for Strategic Cloud Platform Services for 15 years in a row

Someone standing tall in the Gartner Magic Quadrant for 15 long years in a row is no simple deal!!!

Strengths of AWS

- **Largest cloud community** – AWS has built a strong global community of cloud professionals, providing significant opportunities for learning and engagement.
- **Cloud-inspired silicon** – AWS has used its cloud computing experience to develop custom silicon designs, including [AWS Graviton](#), [AWS Inferentia](#), and [AWS Trainium](#), which enable tighter integration between hardware and software, improved power efficiency, and greater control over supply chains.
- **Global scale and operational execution** – AWS's significant share of global cloud market revenue has enabled it to build a larger and more robust network of integration partners than some other providers in this analysis, which in turn helps organizations successfully adopt cloud.



A sample architecture with various components/services

KEY AREAS TO UNDERSTAND...

Edge: This would be the landing zone in the cloud for the traffic flow. This goes back and forth between the public cloud and the vendor (application dev/deploy) devices, user/consumer (using the app) devices. Some sample services used in this layer are given below

NAT, VPN, Direct connect, Shield, Route 53, CDN, WAF, TGW, Firewalls ...

Application: Your single/two/three tier applications (production, non production, partner apps, your dev, uat, puat environments etc) with all the aspects considered like security, load-sharing/balancing etc.

Monitor/Governance: Web Console (various Dashboards like compute, identity access, cloud watch, license manager etc)

REGIONS

- Region – A Physical location around the world with a cluster of data centers. Each group of logical data centers is called as an Availability Zone (AZ). Each AWS Region consists of a minimum of three, isolated, and physically separate AZs within a geographic area. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZs to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.
- AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly.
- AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

As of today, Azure has 60+ regions which is far more than that of AWS (34).

Region
Africa (Cape Town)
Asia Pacific (Hong Kong)
Asia Pacific (Tokyo)
Asia Pacific (Seoul)
Asia Pacific (Osaka)
Asia Pacific (Mumbai)
Asia Pacific (Hyderabad)
Asia Pacific (Singapore)
Asia Pacific (Sydney)
Asia Pacific (Jakarta)
Asia Pacific (Melbourne)
Asia Pacific (Malaysia)
Canada (Central)
Canada West (Calgary)
China (Beijing)
China (Ningxia)
Europe (Frankfurt)
Europe (Zurich)
Europe (Stockholm)
Europe (Milan)
Europe (Spain)
Europe (Ireland)
Europe (London)
Europe (Paris)
Israel (Tel Aviv)
Middle East (UAE)
Middle East (Bahrain)
South America (Sao Paulo)
US East (N. Virginia)
US East (Ohio)
AWS GovCloud (US-East)
AWS GovCloud (US-West)
US West (N. California)
US West (Oregon)



Region Code
af-south-1
ap-east-1
ap-northeast-1
ap-northeast-2
ap-northeast-3
ap-south-1
ap-south-2
ap-southeast-1
ap-southeast-2
ap-southeast-3
ap-southeast-4
ap-southeast-5
ca-central-1
ca-west-1
cn-north-1
cn-northwest-1
eu-central-1
eu-central-2
eu-north-1
eu-south-1
eu-south-2
eu-west-1
eu-west-2
eu-west-3
il-central-1
me-central-1
me-south-1
sa-east-1
us-east-1
us-east-2
us-gov-east-1
us-gov-west-1
us-west-1
us-west-2

AVAILABILITY ZONES

One or more discrete data centers with redundant power, networking, and connectivity in an AWS Region.

AZs give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center.

All AZs in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZs.

All traffic between AZs is encrypted.

The network performance is sufficient to accomplish synchronous replication between AZs.

AZs make partitioning applications for high availability easy. If an application is partitioned across AZs, companies are better isolated and protected from issues such as power outages, lightning strikes, tornadoes, earthquakes, and more. AZs are physically separated by a meaningful distance, many kilometers, from any other AZ, although all are within 100 km (60 miles) of each other..

Azure: More regions, but fewer availability zones per region on average.

AWS: Fewer regions, but more availability zones per region on average.

LOCAL ZONES

Purpose: Provide a subset of AWS services to run workloads closer to end-users in specific geographic areas.

Location: Typically located within metropolitan areas.

Services: Offer a selection of AWS services like EC2, VPC, EBS, and RDS.

Use cases: Ideal for applications that require low latency access to AWS services, such as real-time gaming, video streaming, and IoT.

AWS EDGE LOCATIONS

Purpose: Optimize content delivery and network performance by caching content closer to end-users.

Location: Distributed globally, often in areas with high population density.

Services: Primarily used for caching and accelerating content delivery through Amazon CloudFront.

Use cases: Ideal for applications that require fast content delivery, such as websites, videos, and images

KEY DIFFERENCES:

Purpose: Local zones are for running workloads closer to end-users, while edge locations are for optimizing content delivery.

Services: Local zones offer a subset of AWS services, while edge locations are primarily used for content delivery.

Location: Local zones are typically located within metropolitan areas, while edge locations are distributed globally.

Summary: Local zones are like mini-regions that provide a subset of AWS services for low-latency applications, while edge locations are optimized for delivering content quickly to end-users (primary usecase).

The choice between the two depends on the specific needs and the nature of your application.

OUTPOSTS

AWS Outposts is a fully managed service that extends AWS infrastructure, services, APIs, and tools to on-premises locations:

Run native AWS services locally, such as Amazon EC2, Amazon ECS, Amazon EKS, Amazon EBS, Amazon S3, Amazon RDS, and Amazon EMR.

Get low latency access to on-premises systems for workloads that require it.

Process data locally.

Comply with data residency requirements.

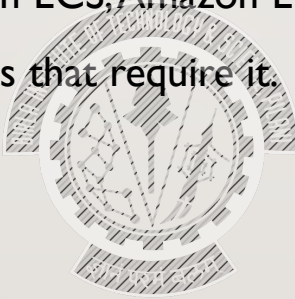
Migrate applications with local system interdependencies.

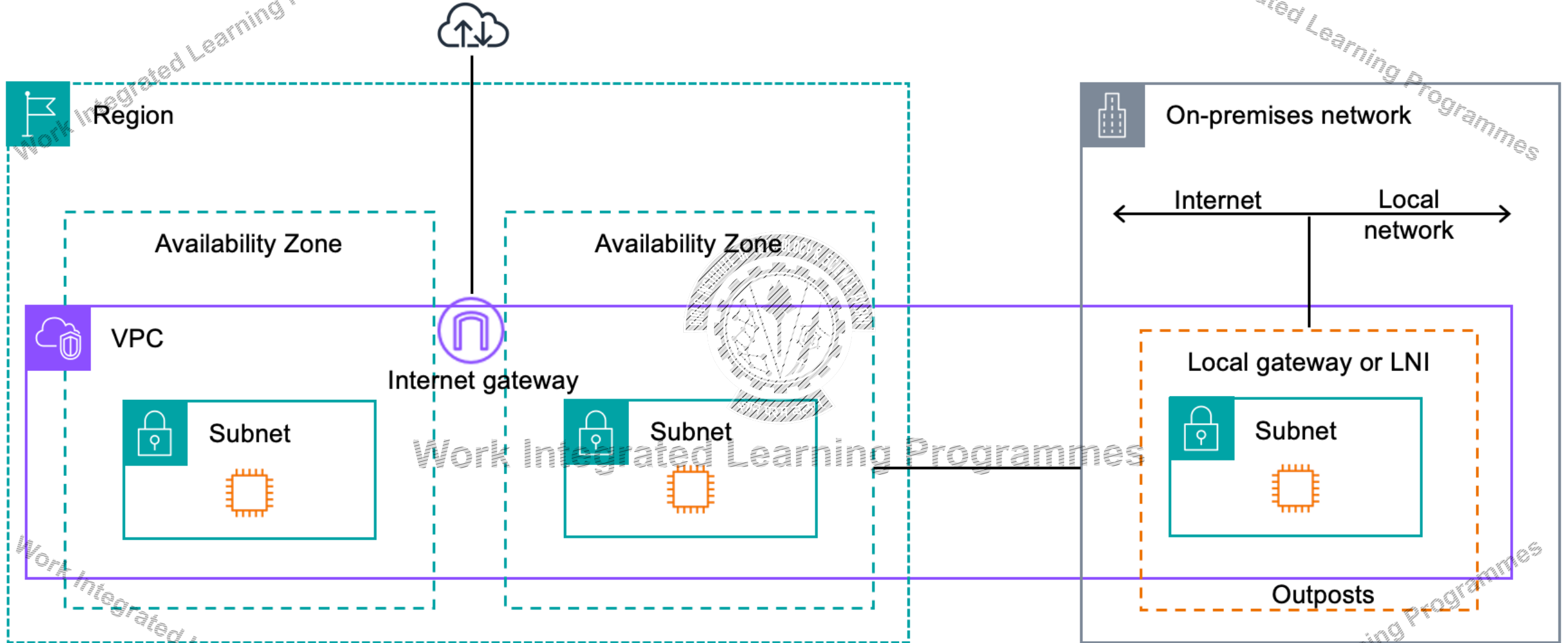
Get a consistent operational experience across on-premises and cloud environments.

AWS manages, monitors, and updates your Outposts infrastructure and AWS services.

Available in a variety of form factors, including 1U and 2U Outposts servers, and 42U Outposts racks.

Outposts are connected to the nearest AWS Region.





AMAZON

Top 25 AWS Services



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AMAZON SERVICES - STORAGE



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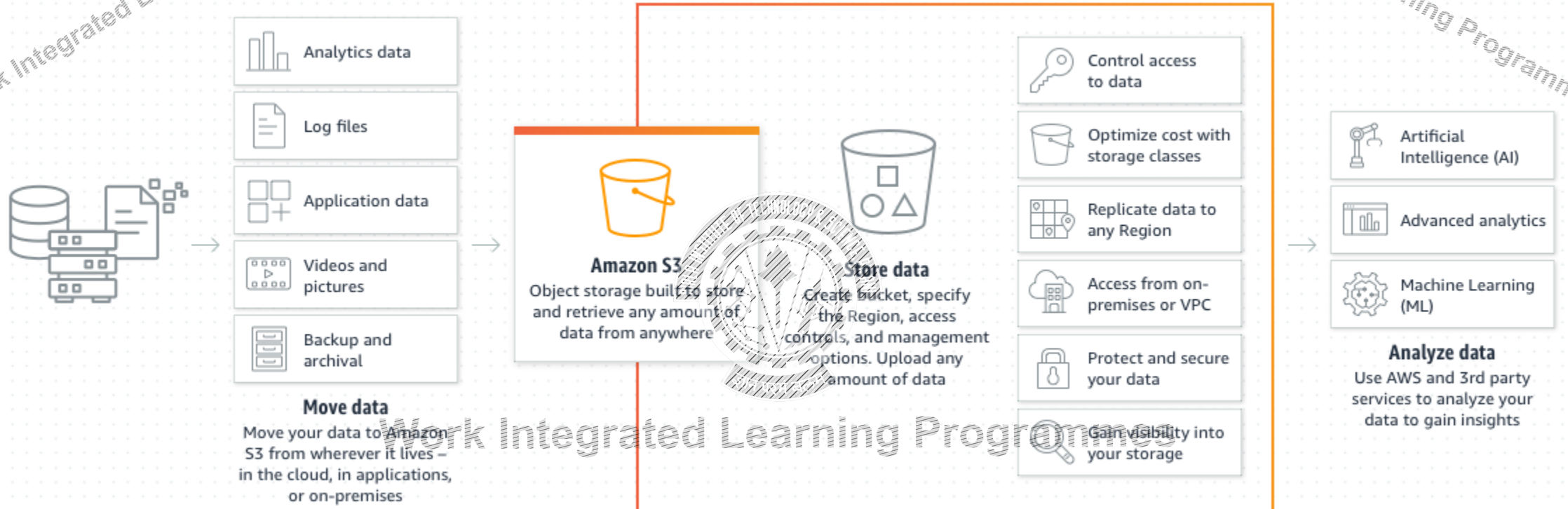
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S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Millions of customers of all sizes and industries store, manage, analyze, and protect any amount of data for virtually any use case, such as data lakes, cloud-native applications, and mobile apps.

With cost-effective storage classes and easy-to-use management features, you can optimize costs, organize and analyze data, and configure fine-tuned access controls to meet specific business and compliance requirements.



GLACIER – STORAGE CLASS OF S3

- Long-term, secure, durable storage classes for data archiving at the lowest cost and milliseconds access
- Purpose-built for data archiving, providing you with the highest performance, most retrieval flexibility, and the lowest cost archive storage in the cloud.
- All S3 Glacier storage classes provide virtually unlimited scalability and are designed for 99.999999999% (11 nines) of data durability.
- The S3 Glacier storage classes deliver options for the fastest access to your archive data and the lowest-cost archive storage in the cloud.

GLACIER – SUB STORAGE CLASSES

Choice of three archive sub storage classes optimized for different access patterns and storage duration.

S3 Glacier Instant Retrieval storage class: For archive data that needs immediate access, such as medical images, news media assets, or genomics data. Delivers the lowest cost storage with milliseconds retrieval.

S3 Glacier Flexible Retrieval storage class: For archive data that does not require immediate access but needs the flexibility to retrieve large sets of data at no cost, such as backup or disaster recovery use cases in minutes or free bulk retrievals in 5-12 hours.

S3 Glacier Deep Archive storage class: To save even more on long-lived archive storage such as compliance archives and digital media preservation, choose, the lowest cost storage in the cloud with data retrieval within twelve hours.

Flexible Retrieval is 10% cheaper than Instant Retrieval – Access is just 2 times a year
Deep Archive is 75% cheaper than Flexible Retrieval – Access less than once per year


Archival data
Medical records, broadcast media, aerial images, consumer photos and videos


Upload directly or use S3 Lifecycle to transition data


Amazon S3 Glacier Instant Retrieval storage class
Milliseconds retrieval of data in a low-cost archive S3 storage class


Amazon S3 Glacier Flexible Retrieval storage class
Minutes to 12 hours retrieval of data in a lower cost archive S3 storage class


Amazon S3 Glacier Deep Archive storage class
12 – 48 hours retrieval of data in the lowest cost archive S3 storage class


Optimize your storage costs with low-cost storage options for long-term digital preservation for rarely accessed data


Ideal for archiving rarely-accessed data — no matter how quickly you need it

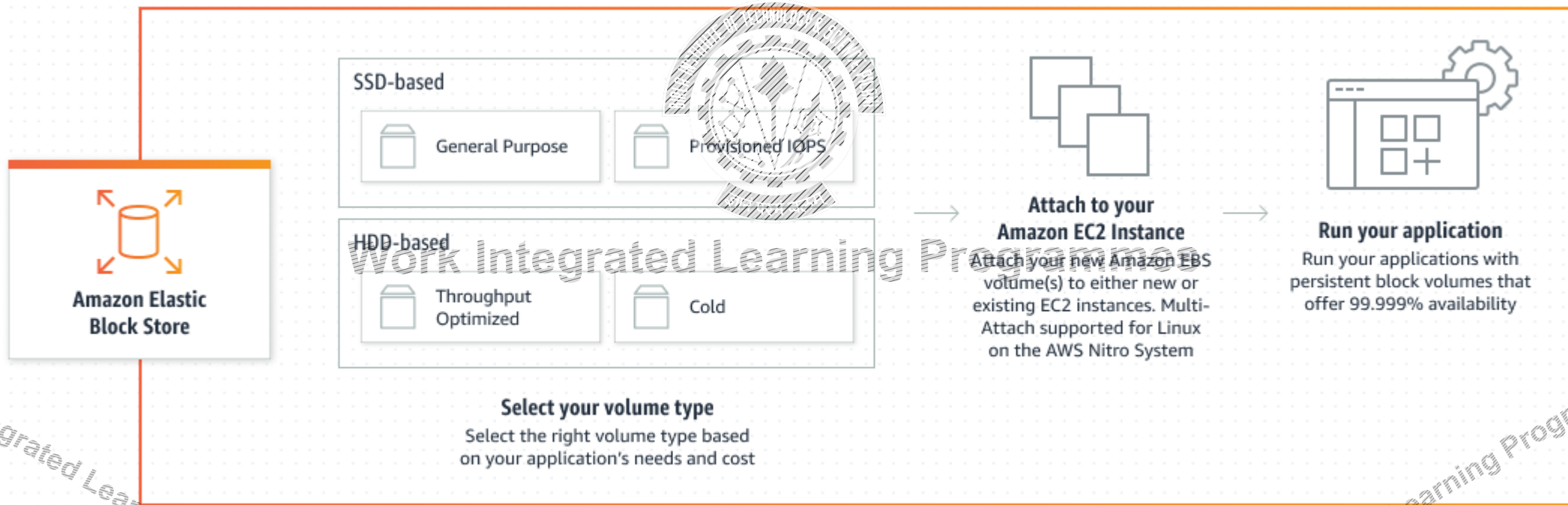

Cost-effective, highly durable, and secure for long-term retention, compliance, and digital preservation


Provides unlimited scale, the highest security standards, and data durability of 11 9s

Increase the value of your digital assets, unlock agility, and save money

ELASTIC BLOCK STORE

Easy to use, high performance block storage at any scale



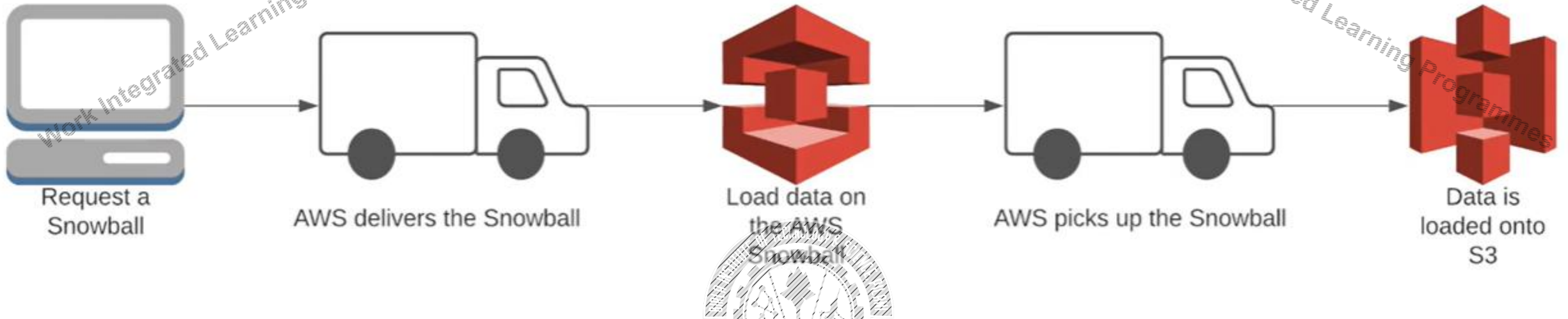
IMPORT/EXPORT SERVICE

AWS Import/Export is a service that allows users to move large amounts of data into and out of Amazon Web Services (AWS) using portable storage devices. It's a faster and more cost-effective alternative to transferring data across the internet, especially for large data sets.

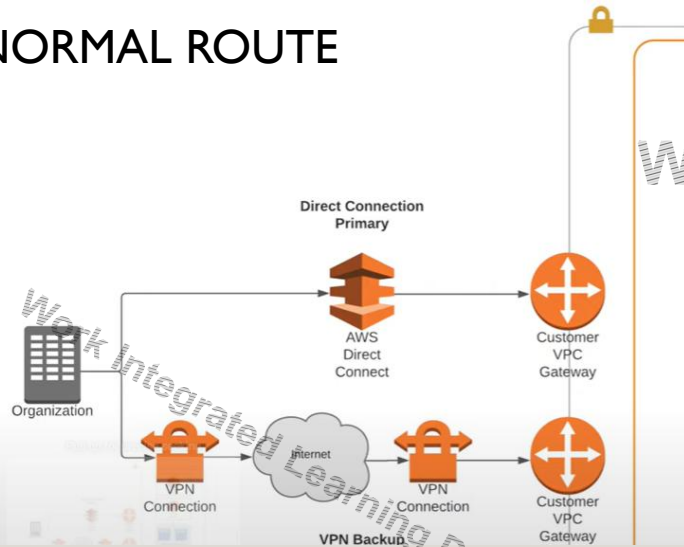


There is also VM export/import. Here are we viewing data export/import

THROUGH SNOWBALL



NORMAL ROUTE



Through Storage Media



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AMAZON SERVICES - DATABASE



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RDS

Easy-to-manage relational database service optimized for total cost of ownership.

Simple to set up, operate, and scale with demand.

Automation for DB management tasks - provisioning, configuring, backups, and patching.

New DB creation – just in minutes

Flexible for customizations - many engines, cloud/onprem, many AZs, read replicas, DR

Performance Optimization - Multi-AZ with two readable standbys, Optimized Writes and Reads, and AWS Graviton3-based instances

Choice of Pricing Options (a typical case for almost all the services)

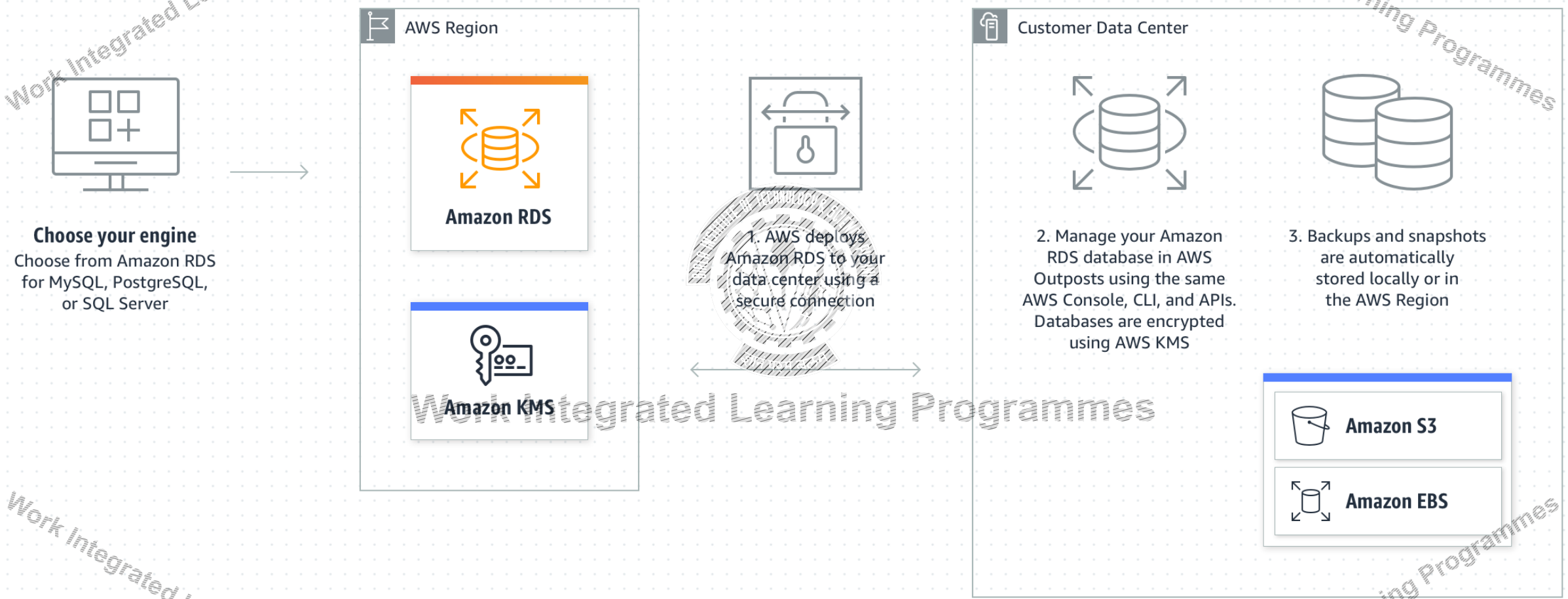
<https://www.youtube.com/watch?v=jW84JhqNWUY>

A Sample Video

RDS - ENGINES

- Amazon Aurora.
- Amazon RDS for PostgreSQL.
- Amazon RDS for MySQL.
- Amazon RDS for MariaDB.
- Amazon RDS for SQL Server.
- Amazon RDS for Oracle.
- Amazon RDS for Db2.





RDS – on prem

DYNAMO DB

Serverless, NoSQL database service for developing modern applications at any scale.

Pay for just what is used

Scales to zero, has no cold starts, no version upgrades, no maintenance windows, no patching, and no downtime maintenance.

DynamoDB offers a broad set of security controls and compliance standards.

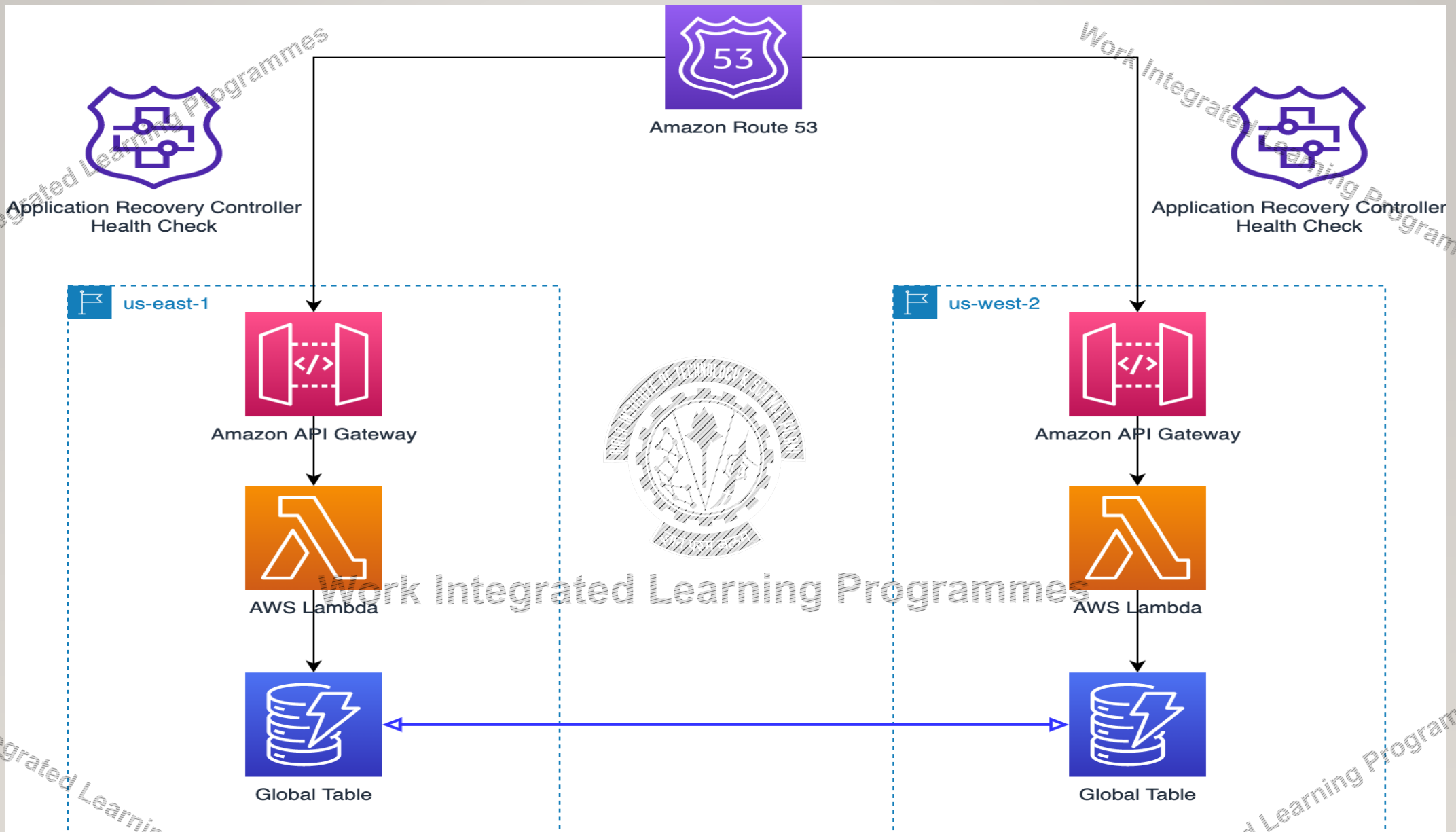
For globally distributed applications, DynamoDB global tables is a multi-Region, multi-active database with a 99.999% availability SLA and increased resilience.

DynamoDB reliability is supported with managed backups, point-in-time recovery, and more.

DynamoDB streams - Build serverless event-driven applications.

<https://youtu.be/sl-zciHAh-4>

Sample Video



A Usage scenario for Dynamo DB Global Tables - Globally distributed Applications

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AMAZON SERVICES – CLOUDFRONT & ELB

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CLOUDFRONT

A content delivery network (CDN) service that helps businesses and developers distribute content to users quickly and reliably.

Uses a worldwide network of data centers called edge locations to deliver content to users with low latency and high data transfer speeds

Designed for security.

Easy to use for developers.

Automated network mapping and intelligent routing to deliver content through its globally dispersed points of presence (POPs).

CloudFront delivers content to users with low latency by routing requests to the edge location with the lowest latency.

Delivers content to users with high transfer speeds

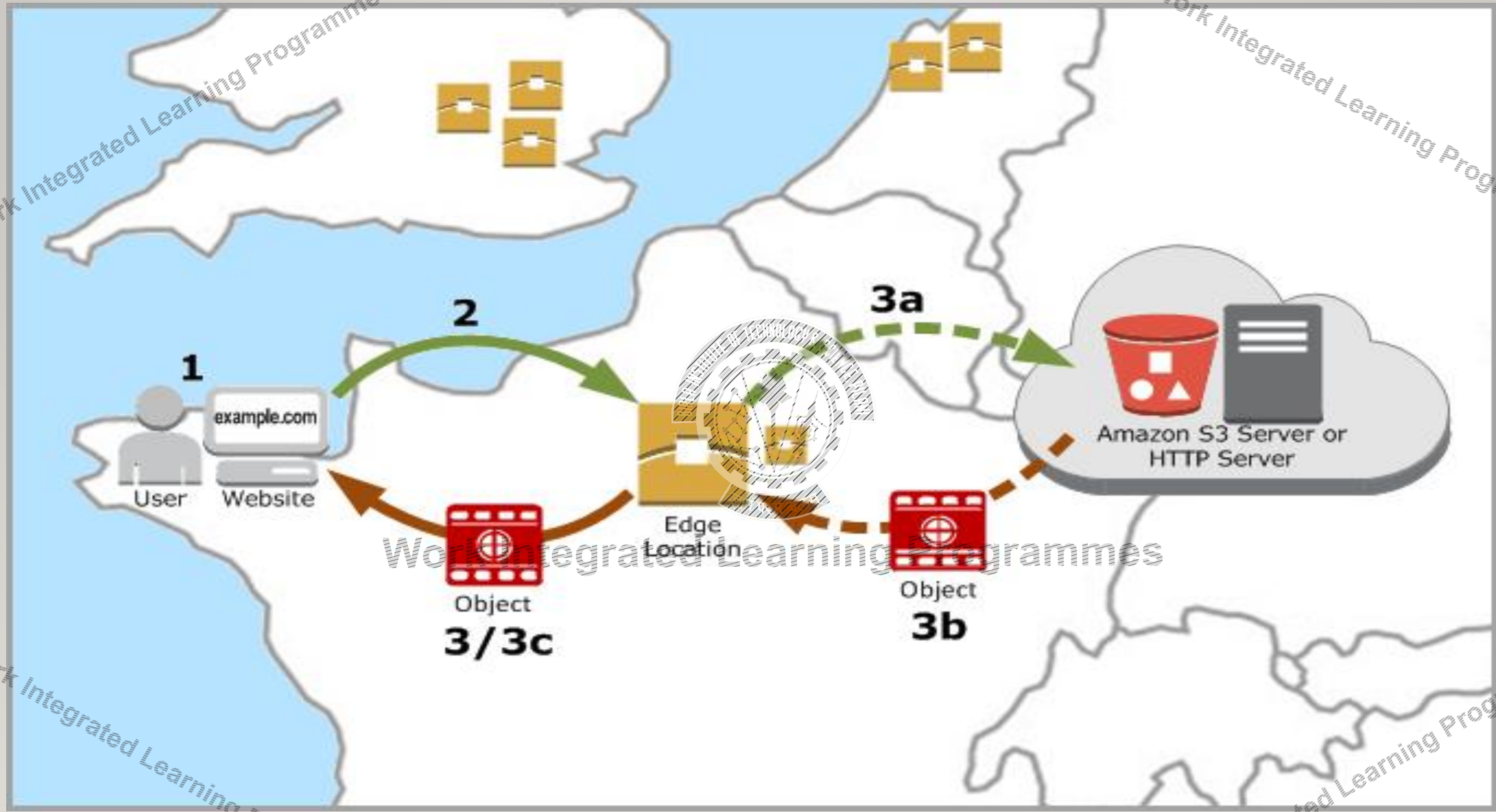


CONTENT DELIVERY

- A user accesses your website or application and sends a request for an object, such as an image file or an HTML file.
- DNS routes the request to the CloudFront POP (edge location) that can best serve the request, typically the nearest CloudFront POP in terms of latency.
- CloudFront checks its cache for the requested object. If the object is in the cache, CloudFront returns it to the user. If the object is *not* in the cache, CloudFront does the following:
 - CloudFront compares the request with the specifications in your distribution and forwards the request to your origin server for the corresponding object—for example, to your Amazon S3 bucket or your HTTP server.
 - The origin server sends the object back to the edge location.
- As soon as the first byte arrives from the origin, CloudFront begins to forward the object to the user. CloudFront also adds the object to the cache for the next time someone requests it.

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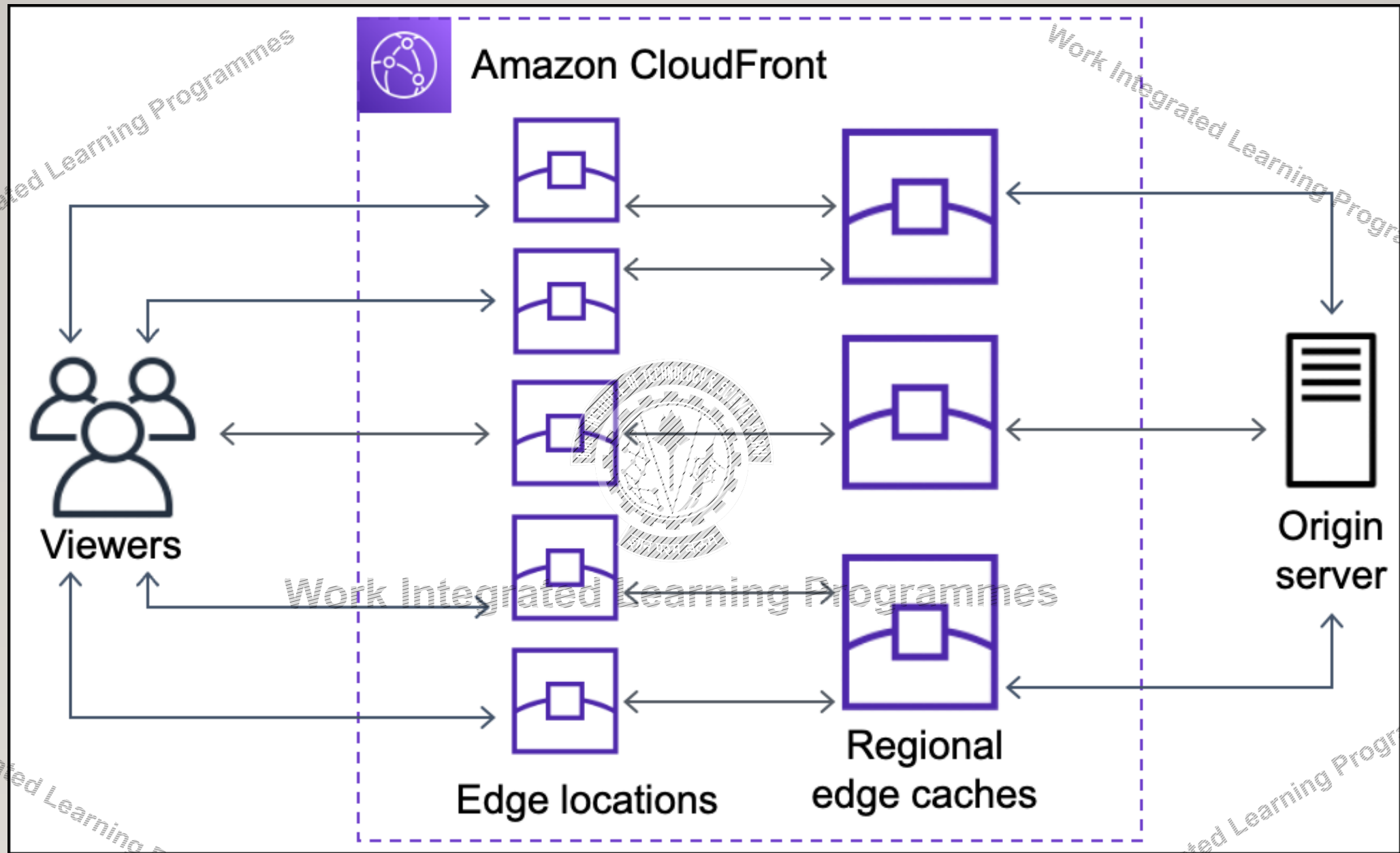
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A Content Accessing Scenario



ELB – ELASTIC LOAD BALANCING

Automatically distributes incoming application traffic across multiple targets and virtual appliances in one or more availability zones.

- Scalability

Automatically scale its request handling capacity based on incoming traffic.

- High availability

ELB can distribute traffic across multiple Availability Zones (AZs) and automatically scale its request handling capacity based on incoming traffic.

- Health checks

ELB runs health checks on targets at a configurable cadence to ensure that they are available and healthy.

- Load balancing

ELB can balance load efficiently between application instances to improve user experience.

